Introduction to Machine Learning Problems: Logistic Regression

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1. (a) i. **Response Variable:** Gender
   ii. **Predictor:** Frequency of audio file
   iii. **Classes:** Male and Female

   (b) i. **Response Variable:** Letter and Number
   ii. **Predictor:** Pressure applied to pad within a given time frame
   iii. **Classes:** 26 Uppercase letters, 26 lowercase letters, and 10 digits

2. (a) $2x_1 + 3x_2 > -1$

   (b)

$$P(y = 1|x) > 0.8 \tag{1}$$
$$\frac{1}{1 + e^{-z}} > 0.8 \tag{2}$$
$$1 + e^{-z} < 1.25 \tag{3}$$
$$z = -ln(0.25) \tag{4}$$
$$= ln(4) \tag{5}$$

   Then, $2x_1 + 3x_2 > ln(4) - 1$

   (c)

$$2x_1 + 3(0.5)) > ln(4) - 1 \tag{6}$$
$$x_1 > \frac{ln(4) - 2.5}{2} \tag{7}$$

3. (a) The data points would be located at $(30, 0)$ and $(70, 1)$ for $y = 0$, and $(50, 1)$, $(80, 2)$, and $(100, 1)$ for $y = 1$.

   (b) $w_1 = 1$, $w_2 = 0$, and $b = -40$. The Linear Classifier is $z_i = x_{i1} - 40$. Sample 3: $70 - 40 = 30 > 0 \implies \hat{y} = 1$ is incorrect.

   (c) Sample 3 has $z_3 = 30$. e^{-30} is very small and thus reduced the probability to effectively 1.

(d) i. The predicted classes would not change because $\alpha > 0$ retains the sign the of equation.

ii. It would push the euler part to 0 faster.

4. (a) $z = -6 + 0.05(40) + 1(3.5) = -6 + 2 + 3.5 = -0.5$, then $P(Y = 1|X) = \frac{1}{1+e^{0.5}}$

(b) $0 = -6 + 0.05x_1 + 3.5$, then $x_1 = 50hours$

5. (a)

$$\frac{\delta z_i}{\delta \beta_0} = 1 \tag{8}$$

$$\frac{\delta z_i}{\delta \beta_1} = x_{1i} \tag{9}$$

$$\frac{\delta z_i}{\delta \beta_2} = x_{2i} \tag{10}$$

(b)

$$\frac{\partial J}{\partial \beta_j} = \sum_{i=1}^{N} \left(\frac{1}{1+e^{z_i}} e^{z_i} \frac{\partial z_i}{\partial \beta_j} - y_i \frac{\partial z_i}{\partial \beta_j} \right) \tag{11}$$

$$\frac{\partial J}{\partial \beta_0} = \sum_{i=1}^{N} \left(\frac{e^{z_i}}{1+e^{z_i}} - y_i \right) \tag{12}$$

$$\frac{\partial J}{\partial \beta_1} = \sum_{i=1}^{N} \left(\frac{e^{z_i}}{1+e^{z_i}} - y_i \right) x_{1i} \tag{13}$$

$$\frac{\partial J}{\partial \beta_2} = \sum_{i=1}^{N} \left(\frac{e^{z_i}}{1+e^{z_i}} - y_i \right) x_{2i} \tag{14}$$

(c) No we can not isolate the β variables. Gradient Descent can be used to determine the slope of the loss function and update the parameters for each step.